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Analysis of ultraviolet and thermal degradations of four common microplastics and evidence of nanoparticle release

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ABSTRACT

Many environmental factors affect the breakdown of plastics in aquatic environments, including exposure to ultraviolet (UV) irradiation and elevated environmental temperatures. More studies are needed to understand how these stressors contribute to plastic degradation, resulting in the release of smaller plastic particles. We studied the impact of environmentally relevant UV and temperature (37 °C) weathering of four high-production volume plastics (polystyrene, polypropylene, low-density polyethylene, and high-density polyethylene) suspended in water. Particle release was detected, characterized by scanning electron microscopy (SEM), and nanoparticles were quantified by nanoparticle tracking analysis (NTA). Weathering resulted in the release of micro- and nanoparticles that exhibited a plastic signature corresponding to the parent microplastic. Nanoparticle release is broadly correlated with an increase in the carbonyl index of the parent microplastic. Aged microplastics were characterized for physical and chemical changes. The impact of weathering on microplastic surface hardness and polymer oxidation depended on material type and environmental factors. Few to no particles were observed in controls, including controls that contained microplastics at 4 °C in dark conditions, highlighting the importance of weathering stimuli in particle release. These results show that plastic degradation needs to consider both the parent microplastic and the smaller particles that are formed.

1. Introduction

Plastics, especially those used for packaging, are materials with long lifespans which are often produced for limited expected use. Polystyrene (PS), polypropylene (PP), and polyethylene (PE) represent the bulk of single-use plastics, (Meides et al., 2021) and their disposal can lead directly (littering) or indirectly (poor process controls) to environmental release (Alimi et al., 2018). Roughly 11% of produced plastic waste was estimated to enter aquatic environments in 2016 (Borrelle et al., 2020). The forecast increase in plastic production and subsequent environmental release is such that reducing pollution levels cannot be achieved by improving waste management streams alone (Borrelle et al., 2020). Additionally, the current environmental load represents an embedded toxicity debt, in which the plastic that is already present will continue to impact the environment as these plastics age, regardless of future inputs (Rillig et al., 2021).

Microplastics (particles >~1000 nm) have been observed worldwide, and more comprehensive data on their occurrence and distribution in natural environments have recently been reported in the literature (Alimi et al., 2018; Nguyen et al., 2019; Barchiesi et al., 2021; Li et al., 2020; Wu et al., 2022; Sarijan et al., 2021). Current sampling methods include beach combing (macroscopic marine debris), biological sampling of wildlife (ingested plastics), investigations of marine sediments, (Coyle et al., 2020) and most commonly, filtration for plastic debris collection from surface water (Nguyen et al., 2019; Wang et al., 2022). These sampling techniques are mainly suitable for detecting microplastics and larger particles, and as a result, there is little information reported on the environmental concentrations of nanoplastics (<~1000 nm) (Gigault et al., 2021; Jakubowicz et al., 2021).

The weathering of microplastics can result in structural changes of polymers (e.g., macromolecular group modification, reduced molecular mass due to chain scission, change in crystallinity content, formation of

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free radicals) (Castro Monsores et al., 2021; McKeen, 2014). UV irradiation can lead to cracking, discoloration, and an increase in the brittleness of the bulk polymer, which has been attributed primarily to oxidation at the surface (Meides et al., 2021; Jakubowicz et al., 2021; Sait et al., 2021; Biber et al., 2019). Thermal degradation, though less explored, has been linked to increases in surface roughness and decreased tensile strength in thin films of high density polyethylene (HDPE) exposed to near-boiling (85–90 °C) water (Ranjan et al., 2021). While environmental water temperatures will rarely reach these extremes, NASA satellite data show surface temperatures of tropical oceans routinely approaching 35 °C (NASA, 2022; University of Hawaii, 2023). Shallow, stagnant bodies of freshwater are likely to experience similar temperatures. In environmental weathering scenarios, processes can work synergistically. For example, UV irradiation that oxidizes polymer chains can increase the crystallinity of the material (notably for PS), rendering it more brittle and enhancing the potential impact of subsequent mechanical abrasion (Castro Monsores et al., 2021). UV coupled to elevated temperatures has been shown to also decrease the resistance of the polymer to thermal decomposition (Luo et al., 2020).

Despite the lack of data on environmental nanoplastic abundance, the release of nanoplastics due to bulk plastic degradation has been observed in laboratory settings following thermal, (Hernandez et al., 2019) ultraviolet light (UV), (Lambert and Wagner, 2016) biological (Kwak and An, 2021) exposures, and mechanical abrasion (Winkler et al., 2022). Ultraviolet irradiation of expanded polystyrene resulted in an appreciable loss of mass and release of micro- and nanoparticles within 3 months in outdoor weathering tests (Song et al., 2020). Similarly, stirred LDPE exposed to very high UVA and UVB irradiance in a solar simulator (1 year equivalent within 3 days) led to microplastic degradation and nanoparticle formation (Sorasan et al., 2021). This latter work agrees with results in which the combination of UV and mechanical shear led to rapid degradation of PE microplastics (Lapointe et al., 2020). In quiescent conditions without UV light, weathering of polyethylene terephthalate and PP thin films at temperatures up to 36.9 °C led to visible pitting of the surfaces in SEM images and the production of micro- and nanoparticles after 620 days (Qin et al., 2022). The addition of a Fenton-like reaction to produce reactive oxygen species led to greater degradation and particle release.

Considerable uncertainty remains regarding the relative impacts of weathering processes across plastic types (e.g., UV versus thermal, PS versus PP). Furthermore, reports generally focus on bulk plastic or microplastics, while few studies aimed to characterize the release of smaller particles as a result of plastic weathering (Alimi et al., 2021). This study examines the release of nanoparticles resulting from the breakdown of four commonly used polymers under two weathering scenarios. Degradation was achieved via laboratory weathering experiments to expose suspended microplastics of PS, PP, HDPE, and low-density polyethylene (LDPE) to either UV irradiation or elevated temperature (37 °C), to understand the relative influence of these environmental factors.

2. Materials and methods

2.1. Weathering microplastics with UV irradiation or thermal stimulus

Plastic sheets of PS, PP, HDPE, and LDPE were ordered from McMaster Carr (see details in SI). While none of the polymers had added dyes, the makeup and specific composition of any additives in the PP, HDPE, and LDPE are unknown. Plastic sheets were cut with clean stainless-steel shears into large microplastics (<5 mm) with dimensions 4.3 mm × 4.3 mm × 1.6 mm (herein termed “prepared microplastics”). All glassware was cleaned with three rinses of RO water followed by three rinses of acetone.

For each plastic-type, six clean borosilicate glass test tubes were filled with 8 mL of filtered RO water. Five prepared microplastics of the same plastic-type were placed in each test tube (Fig. S1) which were covered with borosilicate glass plates to reduce contamination from atmospheric deposition. Additionally, procedural blanks containing only RO water were prepared and placed in the same conditions as the samples to account for any contamination arising from sample preparation or processing.

UV irradiation and high temperature conditions were tested individually to determine their relative impact on the weathering of plastics. For UV treatment, three samples and one procedural blank for each of the 4 plastics were placed into a UV-light chamber. The total UV radiant exposure was 2.50×10^8 J m^{-2} (see SI for details and calculations). For high temperature treatment (T), the same number of samples and blanks were placed in a dark incubator (Labnet, 311 DS Digital Shaking Incubator) set to 37 °C. Exposures were maintained for 18 weeks after which the prepared microplastics were removed and placed in a desiccator until further analysis. The remaining liquid from each test tube (hereafter termed UV-leachate, T-leachate, UV-procedural blank, and T-procedural blank) was characterized for the presence of small microparticles and nanoparticles.

Three control samples (5 prepared microplastics in RO water) of each plastic type were placed in a refrigerator (4 °C) and covered with aluminum foil, herein termed fridge-control. Additionally, a procedural blank was prepared and placed at 4 °C and with no light (termed fridge control-procedural blank). Leachates and prepared microplastics were analyzed after 18 weeks of fridge treatment.

2.2. Analysis of particles in leachates using SEM and NTA

Leachates, controls, and blanks were analyzed in triplicate for the presence of released micro-sized and nano-sized particles by scanning electron microscopy (SEM) and nanoparticle tracking analysis (NTA). Details on sample preparation, SEM analysis, and image analysis are in the SI.

The number of nanoparticles < ~1 μm was determined using nanoparticle tracking analysis (NTA, LM14 instrument with 532 nm green laser, NanoSight Ltd.) by introducing 1 mL of leachate into the NTA chamber. A 60 s video was taken (335 shutter aperture, 0–5 gain) and over 200 particles/run were tracked with the NanoSight software. Runs were repeated three times per triplicate (9 times for each leachate), between which the chamber was cleaned with acetone and water. A fresh sample was used for each run.

2.3. Characterization of the chemical composition of leachates and prepared microplastics

Approximately 3 mL of leachate for each plastic and treatment was drop-cast onto aluminum foil. A droplet was added to the aluminum foil, the droplet was left to dry, and this process was repeated approximately 40 times until an observable film developed on the aluminum foil (Fig. S2). Pristine prepared microplastics, weathered prepared microplastics (UV and temperature), and dried films were characterized using a Spectrum TWO Fourier Transform Infrared Spectrometer (FTIR) instrument with a single-bounce diamond (PerkinElmer) in attenuated total reflection mode (ATR) (see parameters in SI). Each sample was analyzed in triplicate.

2.4. Vickers microhardness of pristine and weathered prepared microplastics

Microhardness measurements (Clark CM100 AT Vickers microhardness instrument) were collected by indenting the surface of each pristine, weathered, and fridge-control prepared microplastic using a pyramid-shaped diamond head (10 gf over 10 s, see details in SI). Ten indentations were conducted for pristine plastic, which was reduced to 5 indentations for weathered plastics due to resource constraints. Reducing the number of measurements was not observed to impact the magnitude of variance in the data.

3. Results and discussion

3.1. Size and number of particles in the leachates

Fig. 1 shows SEM images of dried leachates taken at both low and high magnifications, which span a wide range to better characterize the particles present in the leachate (see SI for details). The particle sizes observed at these two magnifications overlapped and are thus presented separately to avoid conflation. The results of particle sizing measurements (using ImageJ) show that all the leachates were mainly composed of nanoparticles (Table S1). The particles appeared to have irregular morphologies and a high degree of polydispersity.

Size distributions of particles observed in the leachates are shown in Fig. 1, inset and Fig. S3. At both magnifications, the polydispersity of released particle sizes was high. However, the average diameters confirm that observed particles were mainly in the < 1 μm range (Table S1). As such, the number concentration of particles can be approximated using NTA (Fig. 2).

NTA was used to provide an order of magnitude estimate of the particle number concentration for nanoparticles. From these data (Fig. 2), all plastic leachates were observed to contain approximately

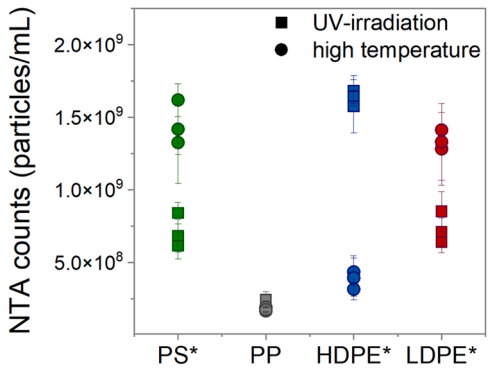


Fig. 2. NTA measurements of the leachates for each treatment. Three samples were measured three times, for a total of 9 measurements per leachate. Asterisks next to polymer type (x-axis) indicate that the average of the 9 measurements for UV-irradiation is significantly different (1-way ANOVA with Tukey's HSD, $\alpha = 0.05$) from the high temperature average. Fridge controls (containing prepared microplastics) and procedural blanks measured by NTA did not contain significant numbers of nanoparticles.

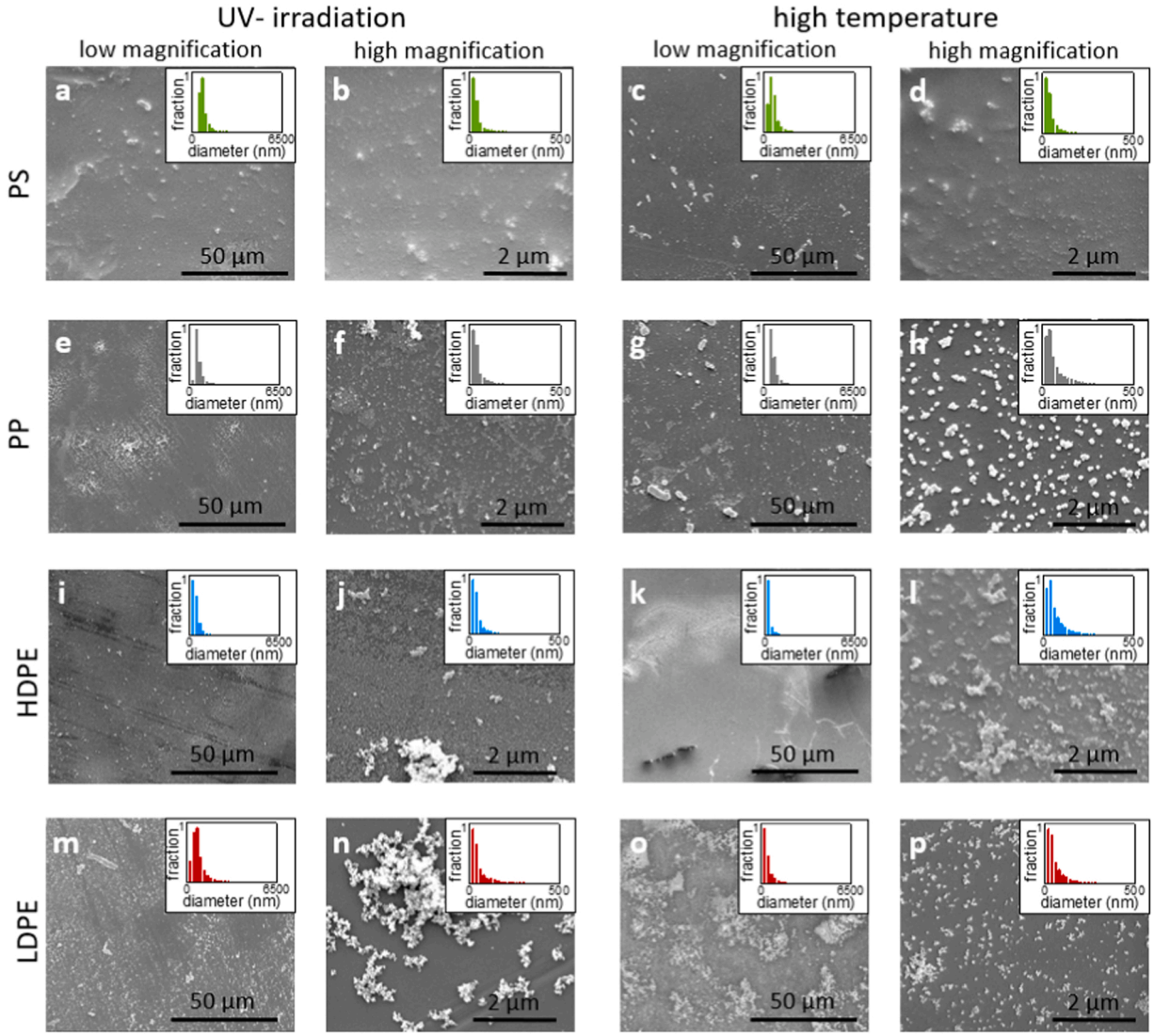


Fig. 1. SEM images of the leachates of plastics exposed to UV-irradiation or high temperature weathering. (a,e,i,m) Imaging at low magnification of leachates of UV-weathered plastics. (b,f,j,n) Imaging at high magnification of leachates of UV-weathered plastics. (c,g,k,o) Imaging at low magnification of leachates of plastics weathered at high temperature. (d,h,l,p) Imaging at high magnification of leachates of plastics weathered at high temperature. All particle size distributions are shown in insets and in Fig. S3.

10^8-10^9 nanoparticles mL^{-1}. HDPE released significantly more nanoparticles when degraded by UV versus high temperature. In contrast, PS and LDPE released significantly more nanoparticles when exposed to high temperature. PP released the fewest number of nanoparticles following weathering, and there was no significant difference between UV and temperature weathering. These results generally agree with the reported stability of PP (Chamas et al., 2020).

3.2. FTIR analysis of the prepared microplastics and their leachates

FTIR analysis performed on thin layers of dried particles prepared from the leachates confirmed that the chemical composition of the released micro- and nanoparticles matched those of the parent bulk materials. In all cases, characteristic reference peaks of the prepared microplastic type were observed in the dried leachates following UV-irradiation and high temperature treatments (Fig. 3, see SI for details).

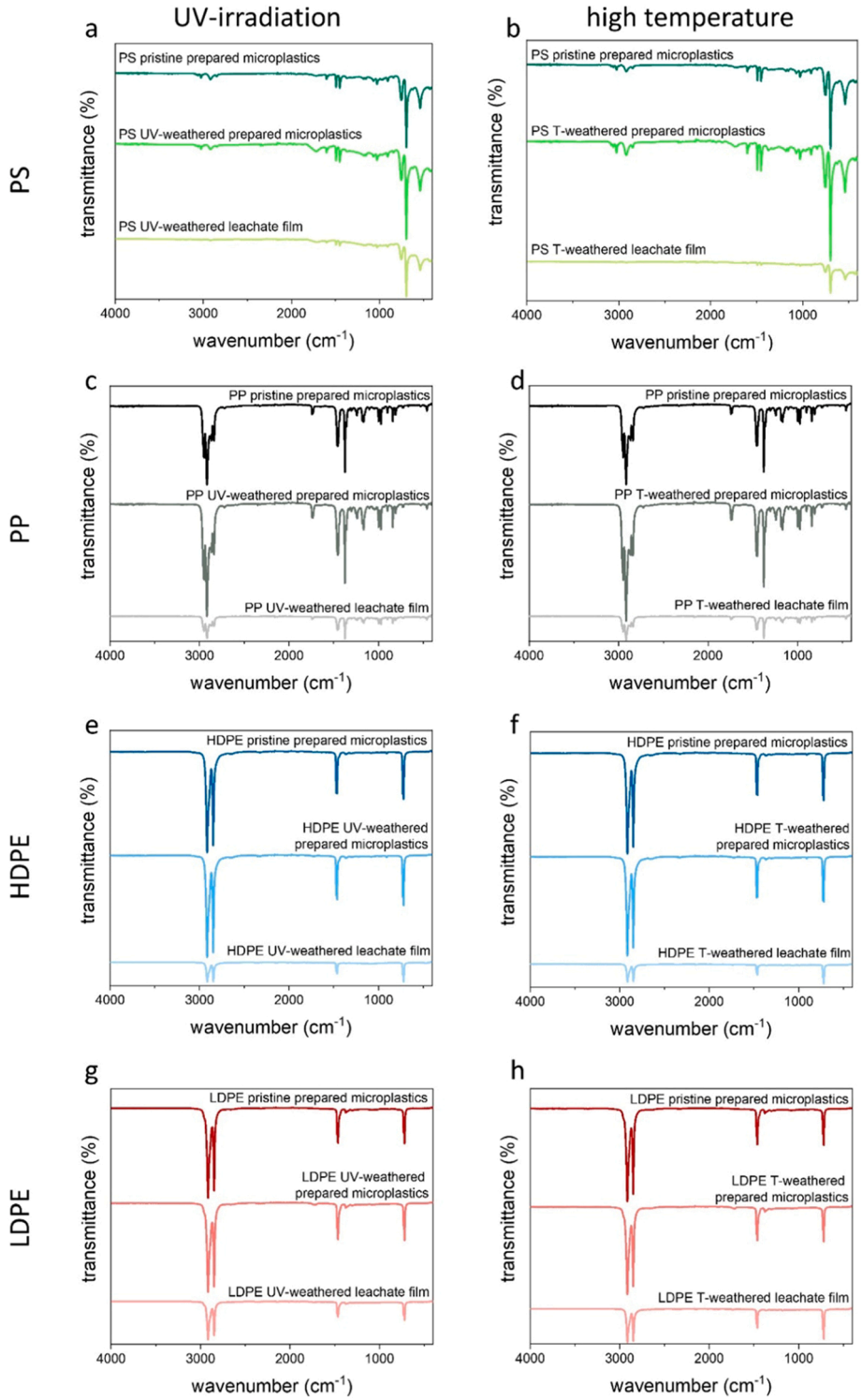


Fig. 3. FTIR spectra for bulk plastics (before and after weathering) and their corresponding films for (a, b) PS, (c, d) PP, (e, f) HDPE, and (g, h) LDPE. The signal for each film corresponds to that of its parent plastic.

Thus, we conclude that each of the plastic types, upon being exposed to UV or thermal treatment for 18 weeks, released plastic particles into the leachate. Drop-casting ~3 mL supernatant from the fridge-control (with prepared microplastics) and procedural blanks (no microplastics) yielded no thin layer for analysis, further confirming that there was no substantial particulate contamination in this study. In general, the absorbance intensity was low in FTIR spectra obtained from drop-cast leachates. Thus, no attempt to analyze the extent of weathering was made using these samples.

Differences in the FTIR spectra between weathered and pristine samples were observed from FTIR measurements performed using prepared microplastics. The formation of carbonyl groups (-C=O, 1550–1815 cm^{-1}) in prepared microplastics confirms oxidation of the weathered surface (Vilaplana et al., 2006). The carbonyl index (CI, Fig. 4, see SI for details) presents a relative measure of oxidation with which to compare the various treatments. The CI increased for PS, HDPE, and LDPE microplastics following both UV- and T-weathering (p-value < 0.1). Only LDPE samples yielded a statistical difference in CI between UV and temperature weathering ($p = 0.047$). No change in CI was observed for PP following either UV- or T-weathering.

3.3. Microhardness changes in prepared microplastics after weathering

Fig. S6 shows the microhardness values for the four different plastics in pristine condition (before weathering), after 18 weeks of either UV-irradiation or high temperature, and after 18 weeks of fridge treatment. The microhardness of the pristine samples decreased as follows: PS > PP > HDPE > LDPE. This can be directly correlated to the different chemical structures of the polyolefins. For instance, PS, bearing phenyl groups, is much more rigid (glass transition temperature $T_{g,PS} \sim 100\ ^{o}C$) than LDPE, consisting solely of branched and aliphatic chains ($T_{g,LDPE}$ typically $< -100\ ^{o}C$) (Greene, 2021).

A marked increase in microhardness from ~13.5–16.0 H_v was measured at the surface of PS after being irradiated with UV for 18 weeks. This result may be explained by a higher degree of crystallinity of UV-exposed PS, as recently reported by Castro Monsores et al. (2021) wherein polystyrene plates exhibited both higher microhardness and higher degrees of crystallinity after being exposed to UV-B (280–320 nm) for 14 days or more. This latter structural change was attributed to crystalline rearrangement after cleavage of the chains. Similar impacts of UV exposure were observed for PS over short time frames by Meides et al., who also observed an increase in elongation at break for PS tensile bars. This was attributed to a temporary increase in polymer degree of crosslinking, although extensibility decreased as weathering continued (Meides et al., 2021).

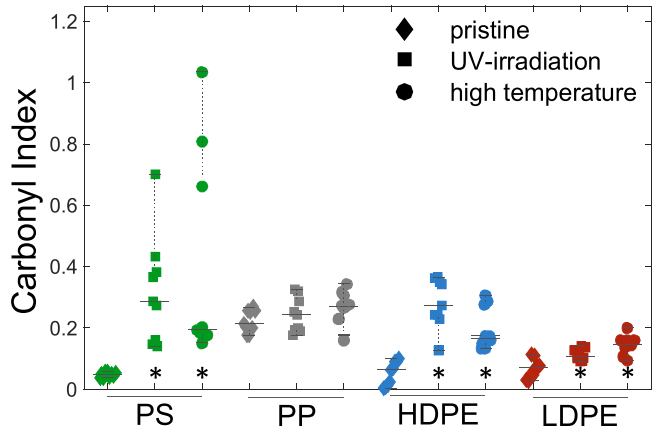


Fig. 4. FTIR carbonyl index calculated from spectra of prepared microplastics for each polymer type and treatment. Solid horizontal lines in the box and whisker plot denotes the median carbonyl index. Asterisks denote statistically significant differences from the pristine sample (1-way ANOVA with Tukey's HSD, $\alpha = 0.1$).

Polypropylene and LDPE microplastics exhibited a decrease in microhardness after being exposed to 37 °C for 18 weeks. Thermolysis and thermo-oxidative degradation at the surface of the plastics are likely causes of LDPE softening. For PP, the CI did not change after weathering, suggesting no evidence of surface oxidation. The decrease in microhardness in PP samples following thermal weathering can then be primarily attributed to thermolysis (random scission followed by radical transfer process), (Saikrasun and Saengsuwan, 2009) which cannot be easily observed by FTIR. As temperature increases, the formation of radicals is accelerated, and the mobility of the chains increases, which favors depolymerization (McKeen, 2014). In principle, we would expect HDPE to also exhibit a decrease in microhardness following exposure to high temperature. However, in this experiment, the results were not significantly different possibly because of heterogeneity within samples or the precision of the instrument.

While the prepared microplastics were floating in RO water during treatment, no visual degradation of the samples by hydrolysis or swelling was observed throughout the study. This is not surprising; all four microplastics are polyolefins which are strongly hydrophobic and particularly resistant to hydrolysis. Thus the timescale for degradation due solely to water exposure is expected to be much longer than examined here. A change in coloration was observed for polystyrene after 18 weeks of UV-irradiation.

Although both temperature and UV play significant roles in plastic degradation, it is not straightforward to establish which of the weathering treatments resulted in greater microplastic degradation. The extent of nanoparticle release appears to be broadly correlated with an increase in the carbonyl index of the original microplastic following weathering (Fig. 4). High temperature weathering produced more LDPE and PS particles when compared to UV-irradiation, while also corresponding to the significant formation of carbonyl groups. In contrast, UV-irradiation produced more HDPE particles when compared to temperature. However, particle release and surface oxidation do not necessarily correspond to the measured surface hardness, as neither UV-irradiated nor high temperature weathered HDPE exhibited any significant difference when compared to the pristine prepared microplastic. Similarly, the microhardness of PS increased significantly when exposed to UV-irradiation but had no significant change when exposed to high temperature. While microhardness is directly related to the bulk molecular structure of the polymers, particle formation is related to polymer chain breakage and the generation of micro-cracks (Meides et al., 2021).

HDPE exhibits a higher melting point (~ 125 °C) than LDPE (~ 110 °C) and is characterized by a higher heat resistance, thus, it should be less susceptible to thermal decomposition. In our study, LDPE produced more particles after temperature weathering than HDPE (Fig. 2, circles). No difference was observed in the CI following temperature weathering between HDPE and LDPE ($\alpha = 0.05$), further suggesting the influence of thermal, rather than oxidative, processes. Conversely, HDPE released more particles and exhibited a greater increase in CI (Fig. 4) than LDPE following UV weathering ($\alpha = 0.05$). Due mainly to the lack of side-chain-branching, HDPE is considered to have greater UV resistance as a bulk characteristic compared to LDPE. This sturdiness of bulk HDPE is observed in the lack of significant change in surface hardness following either UV or temperature weathering, whereas significant differences were observed for LDPE ($\alpha = 0.05$) (Fig. S6).

Other studies, despite different experimental designs, report similar degradation of microplastics that agree with our findings. Biber et al. (2019) observed that degradation, measured as carbonyl content by FTIR, was greater for PS than PE. Furthermore, Meides et al. (2021) observed a multi-stage degradation that initially proceeded slowly as photooxidation occurred in a near-surface layer. Eventually, the formation of micro-cracks and changes to the bulk structure led to an increase in the degradation rate, measured as changes in the average microparticle size (Meides et al., 2021). They observed a decrease in the

polymer number-average molecular weight of PS after 2000 h of UV radiation (from 125,000 g mol^{-1} to 3300 g mol^{-1}) (Meides et al., 2021). This highlighted the occurrence of PS chain scissions at the surface of the samples that resulted in the polymer chain disentanglement and defects and caused embrittlement of the PS (Meides et al., 2021). Recently, Mundhenke et al. (2022) have demonstrated that UV weathering of PE and PP films suspended in ultrapure water is a result of both chain scission and photo-oxidation.

4. Conclusion

We examined the impact of environmentally relevant UV exposure and temperature ageing on large microplastics of PS, PP, LDPE, and HDPE and demonstrate the release of micro- and nanoparticles. This study shows that all four tested plastics release nanoparticles, although differences were observed in the physical (surface hardness) and chemical (CI, FTIR) responses. Plastic degradation should be considered a broad term, as the result of weathering will depend on the considered endpoint (e.g., particle formation, changes in bulk properties), plastic type (e.g., PS, PP, PE), and weathering mechanism (e.g., UV, T). Technology to simultaneously observe and characterize nanoplastics is rapidly being developed, which will allow a more in-depth understanding of the formation dynamics of these small particles. Despite current limitations in detection, nanoplastics already exist in the environment, where their bioavailability and both short- and long-term effects on organisms remain poorly defined. Future research should focus on detailing the mechanism by which these micro- and nanoparticles are released. For example, it remains unclear why changes in CI are the same for UV irradiation and thermal weathering, except for LDPE, in this work. It would be beneficial to determine the molecular response of each plastic to environmental stressors, including but not limited to, UV irradiation and extreme temperatures. Furthermore, while the presence of additives was not observed in this work, we cannot rule out their influence on the plastic responses to weathering in this study, except for PS (100% acrylic copolymer). Future work to detect and quantify additive leaching following weathering may improve our understanding.

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Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.hazl.2023.100078.

References

Alimi, O.S., Farner Budarz, J., Hernandez, L.M., Tufenkji, N., 2018. Microplastics and nanoplastics in aquatic environments: aggregation, deposition, and enhanced contaminant transport. Environ. Sci. Technol. https://doi.org/10.1021/acs.est.7b05559.

Alimi, O.S.; Claveau-Mallet, D.; Kurusu, R.S.; Lapointe, M.; Bayen, S.; Tufenkji, N. Weathering Pathways and Protocols for Environmentally Relevant Microplastics and Nanoplastics: What Are We Missing?, 2021.

Barchiesi, M., Chiavola, A., Di Marcantonio, C., Boni, M.R., 2021. Presence and fate of microplastics in the water sources: focus on the role of wastewater and drinking water treatment plants. J. Water Process Eng. 40, 101787 https://doi.org/10.1016/j.jwpe.2020.101787.

Biber, N.F.A., Foggo, A., Thompson, R.C., 2019. Characterising the deterioration of different plastics in air and seawater. Mar. Pollut. Bull. 141, 595–602. https://doi.org/10.1016/j.marpolbul.2019.02.068.

Borrelle, S.B., Ringma, J., Law, K.L., Monnahan, C.C., Lebreton, L., McGivern, A., Murphy, E., Jambeck, J., Leonard, G.H., Hilleary, M.A., Eriksen, M., Possingham, H. P., De Frond, H., Gerber, L.R., Polidoro, B., Tahir, A., Bernard, M., Mallos, N., Barnes, M., Rochman, C.M., 2020. Predicted growth in plastic waste exceeds efforts to mitigate plastic pollution. Science 369 (6510), 1515–1518. https://doi.org/10.1126/science.aba3656.

Castro Monsores, K.G., de; Silva, A.O., da; Sant' Ana Oliveira, S., de; Weber, R.P., Filho, P.F., Monteiro, S.N., 2021. Influence of ultraviolet radiation on polystyrene. J. Mater. Res. Technol. 13, 359–365. https://doi.org/10.1016/j.jmrt.2021.04.035.

Chamas, A., Moon, H., Zheng, J., Qiu, Y., Tabassum, T., Jang, J.H., Abu-Omar, M., Scott, S.L., Suh, S., 2020. Degradation rates of plastics in the environment. ACS Sustain. Chem. Eng. 8 (9), 3494–3511. https://doi.org/10.1021/acssuschemeng.9b06635.

Coyle, R., Hardiman, G., Driscoll, K.O., 2020. Microplastics in the marine environment: a review of their sources, distribution processes, uptake and exchange in ecosystems. Case Stud. Chem. Environ. Eng. 2, 100010 https://doi.org/10.1016/j.cscee.2020.100010.

Gigault, J., El Hadri, H., Nguyen, B., Grassl, B., Rowenczyk, L., Tufenkji, N., Feng, S., Wiesner, M., 2021. Nanoplastics are neither microplastics nor engineered nanoparticles. Nat. Nanotechnol. 16 (5), 501–507. https://doi.org/10.1038/s41565-021-00886-4.

Greene, J.P., 2021. Microstructures of polymers. Automotive Plastics and Composites. Elsevier,, pp. 27–37. https://doi.org/10.1016/B978-0-12-818008-2.00009-X.

Hernandez, L.M., Xu, E.G., Larsson, H.C.E., Tahara, R., Maisuria, V.B., Tufenkji, N., 2019. Plastic teabags release billions of microparticles and nanoparticles into tea. Environ. Sci. Technol. 53 (21), 12300–12310. https://doi.org/10.1021/acs.est.9b02540.

Jakubowicz, I., Enebro, J., Yarahmadi, N., 2021. Challenges in the search for nanoplastics in the environment—a critical review from the polymer science perspective. Polym. Test. 93, 106953 https://doi.org/10.1016/j.polymertesting.2020.106953.

Kwak, J.I., An, Y.-J., 2021. Microplastic digestion generates fragmented nanoplastics in soils and damages earthworm spermatogenesis and coelomocyte viability. J. Hazard. Mater. 402, 124034 https://doi.org/10.1016/j.jhazmat.2020.124034.

Lambert, S., Wagner, M., 2016. Characterisation of nanoplastics during the degradation of polystyrene. Chemosphere 145, 265–268. https://doi.org/10.1016/j.chemosphere.2015.11.078.

Lapointe, M., Farner, J.M., Hernandez, L.M., Tufenkji, N., 2020. Understanding and improving microplastic removal during water treatment: impact of coagulation and flocculation. Environ. Sci. Technol. 54 (14), 8719–8727. https://doi.org/10.1021/acs.est.0c00712.

Li, C., Busquets, R., Campos, L.C., 2020. Assessment of microplastics in freshwater systems: a review. Sci. Total Environ. 707, 135578 https://doi.org/10.1016/j.scitotenv.2019.135578.

Luo, H., Zhao, Y., Li, Y., Xiang, Y., He, D., Pan, X., 2020. Aging of microplastics affects their surface properties, thermal decomposition, additives leaching and interactions in simulated fluids. Sci. Total Environ. 714, 136862 https://doi.org/10.1016/j.scitotenv.2020.136862.

McKeen, L.W., 2014. Introduction to the effect of heat aging on plastics. The Effect of Long Term Thermal Exposure on Plastics and Elastomers. Elsevier,, pp. 17–42. https://doi.org/10.1016/B978-0-323-22108-5.00002-3.

Meides, N., Menzel, T., Poetzschner, B., Löder, M.G.J., Mansfeld, U., Strohriegl, P., Altstaedt, V., Senker, J., 2021. Reconstructing the environmental degradation of polystyrene by accelerated weathering. Environ. Sci. Technol. 55 (12), 7930–7938. https://doi.org/10.1021/acs.est.0c07718.

Mundhenke, T.F., Li, S.C., Maurer-Jones, M.A., 2022. Photodegradation of polyolefin thin films in simulated freshwater conditions. Environ. Sci.: Process. Impacts 24 (12), 2284–2293. https://doi.org/10.1039/D2EM00359G.

NASA earth observatory. Sea Surface Temperature, 2022. ⟨https://earthobservatory.nasa.gov/global-maps/MYD28M⟩ (accessed 2023–03-27).

Nguyen, B., Claveau-Mallet, D., Hernandez, L.M., Xu, E.G., Farner, J.M., Tufenkji, N., 2019. Separation and analysis of microplastics and nanoplastics in complex environmental samples. Acc. Chem. Res. 52 (4), 858–866. https://doi.org/10.1021/acs.accounts.8b00602.

Qin, J., Zeng, S., Wang, X., Wang, X., Lin, C., 2022. Liberation of plastic nanoparticles and organic compounds from three common plastics in water during weathering under UV radiation-free conditions. Sci. Total Environ. 842, 156859 https://doi.org/10.1016/j.scitotenv.2022.156859.

Ranjan, V.P., Joseph, A., Goel, S., 2021. Microplastics and other harmful substances released from disposable paper cups into hot water. J. Hazard. Mater. 404, 124118 https://doi.org/10.1016/j.jhazmat.2020.124118.

Rillig, M.C., Kim, S.W., Kim, T.-Y., Waldman, W.R., 2021. The global plastic toxicity debt. Environ. Sci. Technol. 55 (5), 2717–2719. https://doi.org/10.1021/acs.est.0c07781.

Saikrasun, S., Saengsuwan, S., 2009. Thermal decomposition kinetics of in situ reinforcing composite based on polypropylene and liquid crystalline polymer. J. Mater. Process. Technol. 209 (7), 3490–3500. https://doi.org/10.1016/j.jmatprotec.2008.08.005.

Sait, S.T.L., Sørensen, L., Kubowicz, S., Vike-Jonas, K., Gonzalez, S.V., Asimakopoulos, A. G., Booth, A.M., 2021. Microplastic fibres from synthetic textiles: environmental degradation and additive chemical content. Environ. Pollut. 268, 115745 https://doi.org/10.1016/j.envpol.2020.115745.

Sarijan, S., Azman, S., Said, M.I.M., Jamal, M.H., 2021. Microplastics in freshwater ecosystems: a recent review of occurrence, analysis, potential impacts, and research needs. Environ. Sci. Pollut. Res 28 (2), 1341–1356. https://doi.org/10.1007/s11356-020-11171-7.

Song, Y.K., Hong, S.H., Eo, S., Han, G.M., Shim, W.J., 2020. Rapid production of micro- and nanoplastics by fragmentation of expanded polystyrene exposed to sunlight. Environ. Sci. Technol. 54 (18), 11191–11200. https://doi.org/10.1021/acs.est.0c02288.

Sorasan, C., Edo, C., González-Pleiter, M., Fernández-Piñas, F., Leganés, F., Rodríguez, A., Rosal, R., 2021. Generation of nanoplastics during the photoageing of low-density polyethylene. Environ. Pollut. 289, 117919 https://doi.org/10.1016/j.envpol.2021.117919.

University of Hawai. Ocean Temperature Profiles. ⟨https://manoa.hawaii.edu/exploringourfluidearth/physical/density-effects/ocean-temperature-profiles⟩ (accessed 2023-03–27).

Vilaplana, F., Ribes-Greus, A., Karlsson, S., 2006. Degradation of recycled high-impact polystyrene. simulation by reprocessing and thermo-oxidation. Polym. Degrad. Stab. 91 (9), 2163–2170. https://doi.org/10.1016/j.polymdegradstab.2006.01.007.

Wang, C., O'Connor, D., Wang, L., Wu, W.-M., Luo, J., Hou, D., 2022. Microplastics in urban runoff: global occurrence and fate. Water Res. 225, 119129 https://doi.org/10.1016/j.watres.2022.119129.

Winkler, A., Fumagalli, F., Cella, C., Gilliland, D., Tremolada, P., Valsesia, A., 2022. Detection and formation mechanisms of secondary nanoplastic released from drinking water bottles. Water Res. 222, 118848 https://doi.org/10.1016/j.watres.2022.118848.

Wu, C., Xiong, X., Hamidian, A.H., Zhang, Y., Xu, X., 2022. A review on source, occurrence, and impacts of microplastics in freshwater aquaculture systems in China. Water Biol. Secur. 1 (3), 100040 https://doi.org/10.1016/j.watbs.2022.100040.